

Observed Variable Model Identification

Agenda

- Observed variable models (a simplification)
- Implied covariance
- Identification with a simple model
- Identification Rules
- Summary

SEMs with Observed Variables

- No latent variables in the model (L)
- No measurement errors
 - Note: ϵ can capture some measurement error but it is confounded with unexplained variance

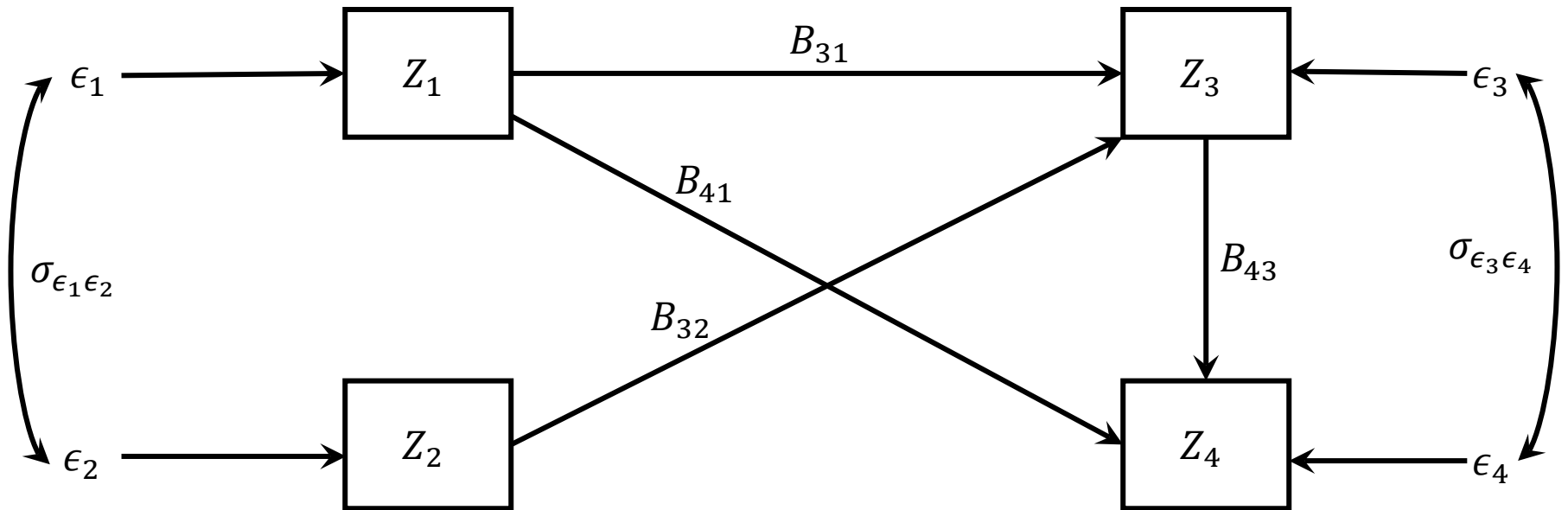
$$\mathbf{Z} = \boldsymbol{\alpha} + \mathbf{B}\mathbf{Z} + \boldsymbol{\epsilon}$$

$\mathbf{B} = N_Z \times N_Z$ coefficient matrix

$\mathbf{Z} = N_Z \times 1$ vector of observed variables

$\boldsymbol{\epsilon} = N_Z \times 1$ vector of errors for each equation

An Example Model



$$\mathbf{Z}_1 = \alpha_1 + \epsilon_1$$

$$\mathbf{Z}_2 = \alpha_2 + \epsilon_2$$

$$\mathbf{Z}_3 = \alpha_3 + B_{31}\mathbf{Z}_1 + B_{32}\mathbf{Z}_2 + \epsilon_3$$

$$\mathbf{Z}_4 = \alpha_4 + B_{41}\mathbf{Z}_1 + B_{42}\mathbf{Z}_2 + B_{43}\mathbf{Z}_3 + \epsilon_4$$

$$\Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_1}^2 & \sigma_{\epsilon_1\epsilon_2} & 0 & 0 \\ \sigma_{\epsilon_1\epsilon_2} & \sigma_{\epsilon_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_3}^2 & \sigma_{\epsilon_3\epsilon_4} \\ 0 & 0 & \sigma_{\epsilon_3\epsilon_4} & \sigma_{\epsilon_4}^2 \end{bmatrix}$$

Recursive Models

We've discussed recursive models when we did path tracing.

Two conditions to pay attention to for observed variable models:

1. $\mathbf{B}$ can be expressed as lower triangular
2. $\Sigma_{\epsilon_Y \epsilon_Y}$ is diagonal

If $\Sigma_{\epsilon_Y \epsilon_Y}$ is not diagonal then we call the model *partially recursive*

Is our example model recursive?

$$\mathbf{Z}_1 = \alpha_1 + \epsilon_1$$

$$\mathbf{Z}_2 = \alpha_2 + \epsilon_2$$

$$\mathbf{Z}_3 = \alpha_3 + B_{31}\mathbf{Z}_1 + B_{32}\mathbf{Z}_2 + \epsilon_3$$

$$\mathbf{Z}_4 = \alpha_4 + B_{41}\mathbf{Z}_1 + B_{43}\mathbf{Z}_3 + \epsilon_4$$

$$\Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_1}^2 & \sigma_{\epsilon_1\epsilon_2} & 0 & 0 \\ \sigma_{\epsilon_1\epsilon_2} & \sigma_{\epsilon_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_3}^2 & \sigma_{\epsilon_3\epsilon_4} \\ 0 & 0 & \sigma_{\epsilon_3\epsilon_4} & \sigma_{\epsilon_4}^2 \end{bmatrix}$$

What is our $\mathbf{B}$ matrix?

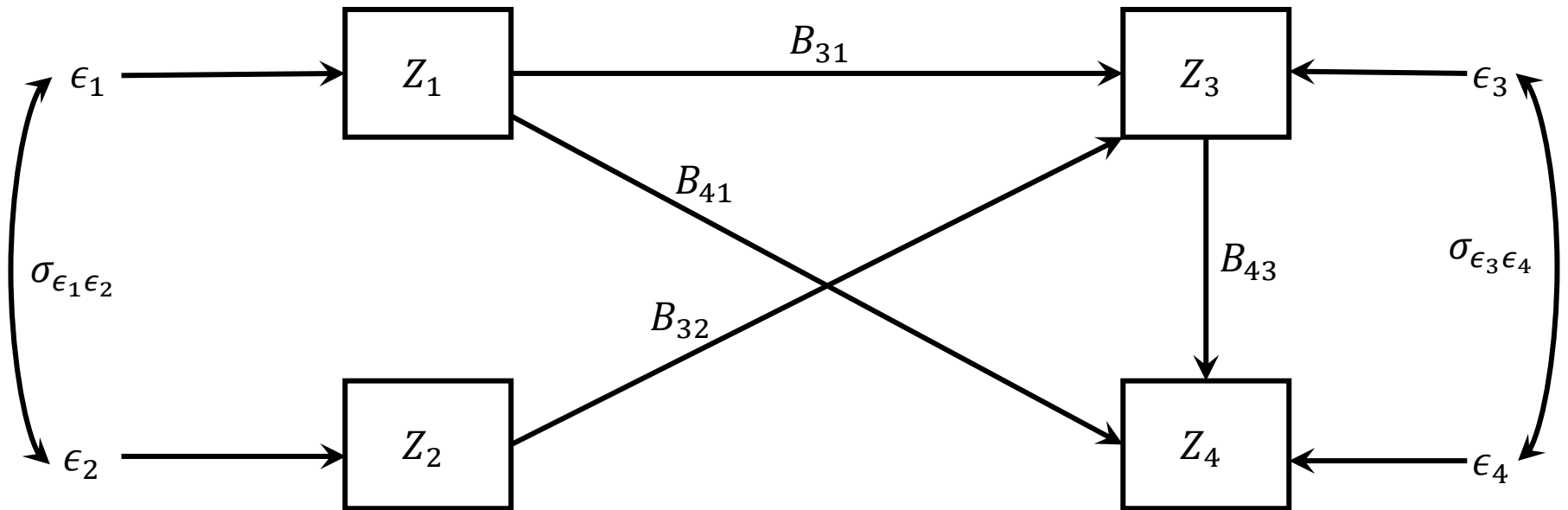
$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ B_{31} & B_{32} & 0 & 0 \\ B_{41} & 0 & B_{43} & 0 \end{bmatrix} \quad \text{Lower triangular, so we're good!}$$

What about $\Sigma_{\epsilon_Y\epsilon_Y}$?

$$\Sigma_{\epsilon_Y\epsilon_Y} = \begin{bmatrix} \sigma_{\epsilon_3}^2 & \sigma_{\epsilon_3\epsilon_4} \\ \sigma_{\epsilon_3\epsilon_4} & \sigma_{\epsilon_4}^2 \end{bmatrix} \quad \text{Not diagonal.}$$

The model is **partially recursive**. How could we change the model to be **recursive**?

An Example *Recursive* Model



$$\mathbf{Z}_1 = \alpha_1 + \epsilon_1$$

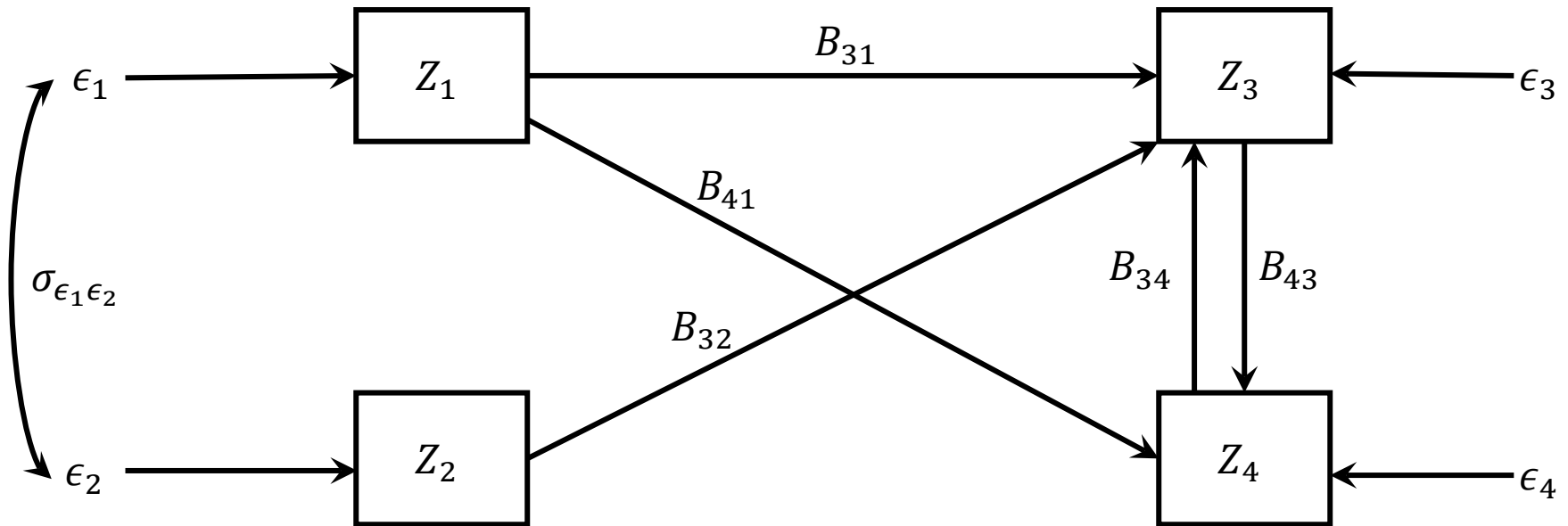
$$\mathbf{Z}_2 = \alpha_2 + \epsilon_2$$

$$\mathbf{Z}_3 = \alpha_3 + B_{31}\mathbf{Z}_1 + B_{32}\mathbf{Z}_2 + \epsilon_3$$

$$\mathbf{Z}_4 = \alpha_4 + B_{41}\mathbf{Z}_1 + B_{43}\mathbf{Z}_3 + \epsilon_4$$

$$\Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_1}^2 & \sigma_{\epsilon_1 \epsilon_2} & 0 & 0 \\ \sigma_{\epsilon_1 \epsilon_2} & \sigma_{\epsilon_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_3}^2 & 0 \\ 0 & 0 & 0 & \sigma_{\epsilon_4}^2 \end{bmatrix}$$

How could we make our model non-recursive?



$$Z_1 = \alpha_1 + \epsilon_1$$

$$Z_2 = \alpha_2 + \epsilon_2$$

$$Z_3 = \alpha_3 + B_{31}Z_1 + B_{32}Z_2 + \epsilon_3$$

$$Z_4 = \alpha_4 + B_{41}Z_1 + B_{43}Z_3 + \epsilon_4$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ B_{31} & B_{32} & 0 & 0 \\ B_{41} & 0 & B_{43} & 0 \end{bmatrix}$$

Implied Moments and Covariance

Implied Means

We can calculate the model implied means

$$\mu(\theta) = (I - B)^{-1}\alpha$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ B_{31} & B_{32} & 0 & 0 \\ B_{41} & 0 & B_{43} & 0 \end{bmatrix}$$

$$(I - B) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -B_{31} & -B_{32} & 1 & 0 \\ -B_{41} & 0 & -B_{43} & 1 \end{bmatrix}$$

$$(I - B)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ B_{31} & B_{32} & 1 & 0 \\ B_{31}B_{43} + B_{41} & B_{32}B_{43} & B_{43} & 1 \end{bmatrix}$$

Implied Means

We can calculate the model implied means

$$\mu(\theta) = (I - B)^{-1}\alpha$$

$$(I - B)^{-1}\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 + B_{31}\alpha_1 + B_{32}\alpha_2 \\ \alpha_4 + (B_{31}B_{43} + B_{41})\alpha_1 + B_{32}B_{43}\alpha_2 + B_{43}\alpha_3 \end{bmatrix}$$

Predicted mean of Z_1 through Z_4

Insights:

- α_1 and α_2 are the predicted means of Z_1 and Z_2
- Predicted mean of Z_3 depends on intercepts and slopes of upstream variables
- Predicted mean of Z_4 depends on intercept of Z_2 , even without direct path

Implied Covariance

We can calculate the model implied covariances

$$\Sigma(\theta) = (I - B)^{-1} \Sigma_{\epsilon\epsilon} (I - B)^{-1'}$$

$$(I - B)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ B_{31} & B_{32} & 1 & 0 \\ B_{31}B_{43} + B_{41} & B_{32}B_{43} & B_{43} & 1 \end{bmatrix} \quad \Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_1}^2 & \sigma_{\epsilon_1\epsilon_2} & 0 & 0 \\ \sigma_{\epsilon_1\epsilon_2} & \sigma_{\epsilon_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_3}^2 & \sigma_{\epsilon_3\epsilon_4} \\ 0 & 0 & \sigma_{\epsilon_3\epsilon_4} & \sigma_{\epsilon_4}^2 \end{bmatrix}$$

$$(I - B)^{-1} \Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_1}^2 & \sigma_{\epsilon_1\epsilon_2} & 0 & 0 \\ \sigma_{\epsilon_1\epsilon_2} & \sigma_{\epsilon_2}^2 & 0 & 0 \\ B_{31}\sigma_{\epsilon_1}^2 + B_{32}\sigma_{\epsilon_1\epsilon_2} & B_{31}\sigma_{\epsilon_1\epsilon_2} + B_{32}\sigma_{\epsilon_2}^2 & \sigma_{\epsilon_3}^2 & \sigma_{\epsilon_3\epsilon_4} \\ (B_{31}B_{43} + B_{41})\sigma_{\epsilon_1}^2 + B_{32}B_{43}\sigma_{\epsilon_1\epsilon_2} & (B_{31}B_{43} + B_{41})\sigma_{\epsilon_1\epsilon_2} + B_{32}B_{43}\sigma_{\epsilon_2}^2 & B_{43}\sigma_{\epsilon_3\epsilon_4} & B_{43}\sigma_{\epsilon_3\epsilon_4} + \sigma_{\epsilon_4}^2 \end{bmatrix}$$

Implied Covariance

We can calculate the model implied covariances

$$\Sigma(\boldsymbol{\theta}) = (\mathbf{I} - \mathbf{B})^{-1} \Sigma_{\epsilon\epsilon} (\mathbf{I} - \mathbf{B})^{-1}$$

$$\sigma_{Z_1}^2(\boldsymbol{\theta}) = \sigma_{\epsilon_1}^2$$

$$\sigma_{Z_1 Z_2}(\boldsymbol{\theta}) = \sigma_{\epsilon_1 \epsilon_2} \qquad \sigma_{Z_2}^2(\boldsymbol{\theta}) = \sigma_{\epsilon_2}^2$$

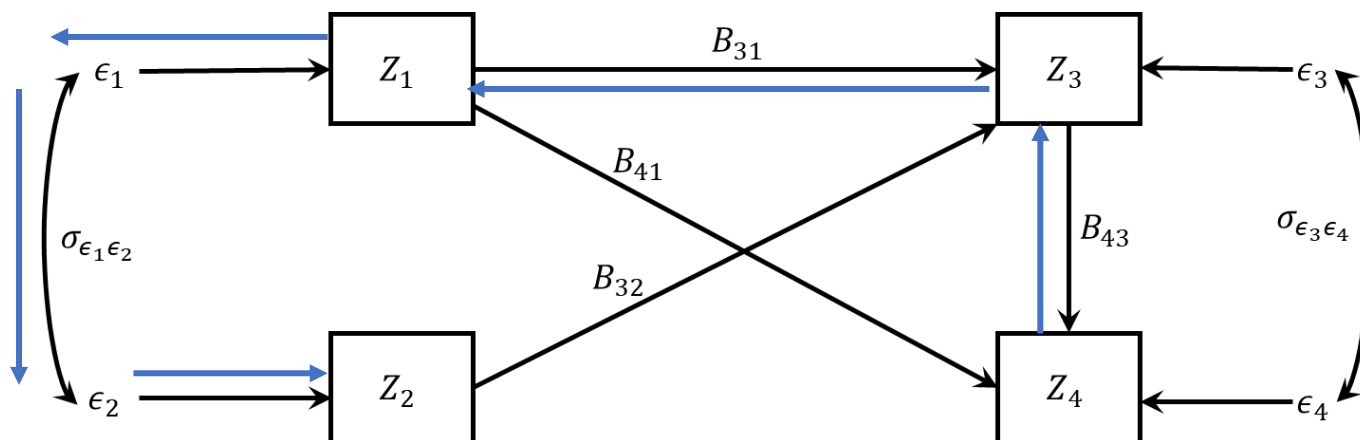
$$\sigma_{Z_1 Z_3}(\boldsymbol{\theta}) = B_{31} \sigma_{\epsilon_1}^2 + B_{32} \sigma_{\epsilon_1 \epsilon_2} + B_{31} \sigma_{\epsilon_3}^2 + (B_{31} B_{43} + B_{41}) \sigma_{\epsilon_3 \epsilon_4}$$

$$\begin{aligned} \sigma_{Z_1 Z_4}(\boldsymbol{\theta}) = & (B_{31} B_{43} + B_{41}) \sigma_{\epsilon_1}^2 + B_{32} B_{43} \sigma_{\epsilon_1 \epsilon_2} + B_{31} B_{43} \sigma_{\epsilon_3 \epsilon_4} \\ & + (B_{31} B_{43} + B_{41}) (B_{43} \sigma_{\epsilon_3 \epsilon_4} + \sigma_{\epsilon_4}^2) \end{aligned}$$

It may look like non-sense, but there are insights to see.

Covariance algebra aligns with path tracing

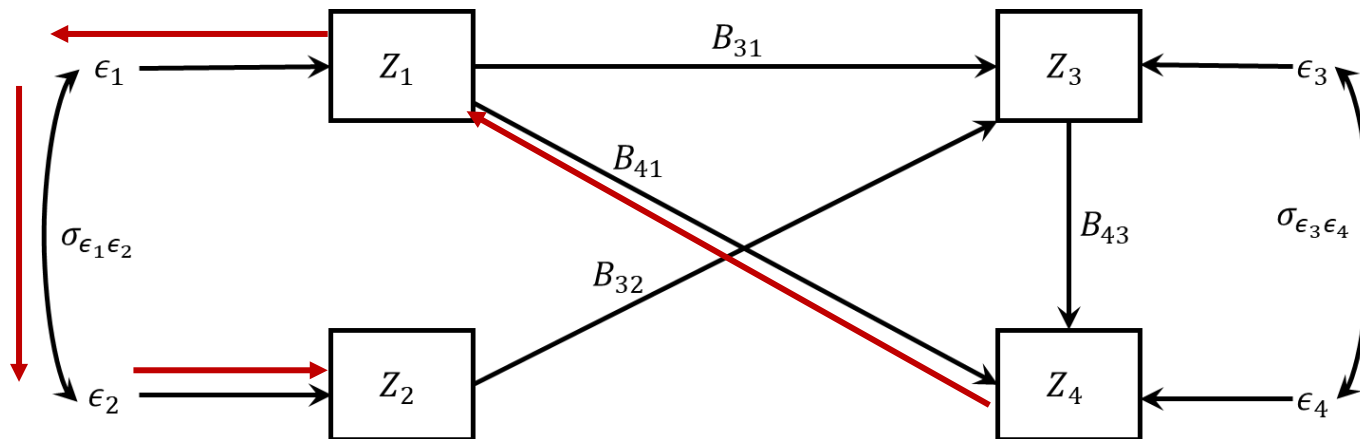
Examining $\sigma_{Z_2Z_4}(\theta)$



There is no direct path from Z_2 to Z_4 but that does not mean they do not covary

$$\begin{aligned}\sigma_{Z_2Z_4}(\theta) &= (B_{31}B_{43} + B_{41})\sigma_{\epsilon_1\epsilon_2} + B_{32}B_{43}\sigma_{\epsilon_2}^2 \\ &= \boxed{B_{43}B_{31}\sigma_{\epsilon_1\epsilon_2}} + B_{41}\sigma_{\epsilon_1\epsilon_2} + B_{32}B_{43}\sigma_{\epsilon_2}^2\end{aligned}$$

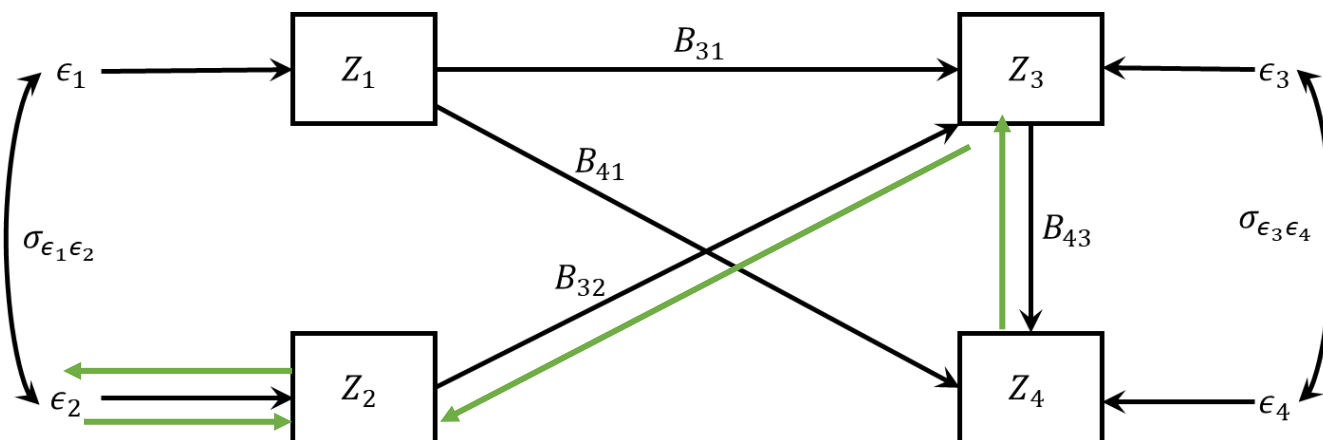
Examining $\sigma_{Z_2Z_4}(\theta)$



There is no direct path from Z_2 to Z_4 but that does not mean they do not covary

$$\begin{aligned}\sigma_{Z_2Z_4}(\theta) &= (B_{31}B_{43} + B_{41})\sigma_{\epsilon_1\epsilon_2} + B_{32}B_{43}\sigma_{\epsilon_2}^2 \\ &= \boxed{B_{43}B_{31}\sigma_{\epsilon_1\epsilon_2}} + \boxed{B_{41}\sigma_{\epsilon_1\epsilon_2}} + B_{32}B_{43}\sigma_{\epsilon_2}^2\end{aligned}$$

Examining $\sigma_{Z_2 Z_4}(\theta)$



There is no direct path from Z_2 to Z_4 but that does not mean they do not covary

$$\begin{aligned}\sigma_{Z_2 Z_4}(\theta) &= (B_{31}B_{43} + B_{41})\sigma_{\epsilon_1 \epsilon_2} + B_{32}B_{43}\sigma_{\epsilon_2}^2 \\ &= \boxed{B_{43}B_{31}\sigma_{\epsilon_1 \epsilon_2}} + \boxed{B_{41}\sigma_{\epsilon_1 \epsilon_2}} + \boxed{B_{43}B_{32}\sigma_{\epsilon_2}^2}\end{aligned}$$

Z_2 and Z_4 covary for 3 reasons:

1. Z_2 covaries with Z_1 which affects Z_3 which then affects Z_4 (indirect effect)
2. Z_2 covaries with Z_1 which affects Z_4 directly
3. Z_2 affects Z_3 which then affects Z_4 (indirect effect)

Covariance at Population vs. Model

Σ_{ZZ} is the population covariance matrix of all the Z s

$$\Sigma_{ZZ} = \left[\begin{array}{c|c} \Sigma_{XX} & \Sigma_{YX} \\ \hline \Sigma_{XY} & \Sigma_{YY} \end{array} \right]$$

We will get an estimate of Σ_{ZZ} from our data

The model will generate an estimate of Σ_{ZZ} , which we call $\Sigma_{ZZ}(\theta)$

θ is a vector of all freely estimated parameters in the model

$$\Sigma_{ZZ}(\theta) = \left[\begin{array}{c|c} \Sigma_{XX}(\theta) & \Sigma_{YX}(\theta) \\ \hline \Sigma_{YX}(\theta) & \Sigma_{YY}(\theta) \end{array} \right]$$

Estimated Parameters

Consider our partially recursive example

$$\begin{aligned}
 \mathbf{Z}_1 &= \alpha_1 + \epsilon_1 \\
 \mathbf{Z}_2 &= \alpha_2 + \epsilon_2 \\
 \mathbf{Z}_3 &= \alpha_3 + B_{31}\mathbf{Z}_1 + B_{32}\mathbf{Z}_2 + \epsilon_3 \\
 \mathbf{Z}_4 &= \alpha_4 + B_{41}\mathbf{Z}_1 + B_{43}\mathbf{Z}_3 + \epsilon_4
 \end{aligned}
 \quad
 \Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_1}^2 & \sigma_{\epsilon_1\epsilon_2} & 0 & 0 \\ \sigma_{\epsilon_1\epsilon_2} & \sigma_{\epsilon_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_3}^2 & \sigma_{\epsilon_3\epsilon_4} \\ 0 & 0 & \sigma_{\epsilon_3\epsilon_4} & \sigma_{\epsilon_4}^2 \end{bmatrix}$$

All the estimated parameters:

$$\theta_{\mu,\Sigma} = [\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \quad B_{31} \quad B_{32} \quad B_{41} \quad B_{43} \quad \sigma_{\epsilon_1}^2 \quad \sigma_{\epsilon_2}^2 \quad \sigma_{\epsilon_3}^2 \quad \sigma_{\epsilon_4}^2 \quad \sigma_{\epsilon_1\epsilon_2} \quad \sigma_{\epsilon_3\epsilon_4}]$$

For now we will ignore the intercepts

$$\theta_{\Sigma} = [B_{31} \quad B_{32} \quad B_{41} \quad B_{43} \quad \sigma_{\epsilon_1}^2 \quad \sigma_{\epsilon_2}^2 \quad \sigma_{\epsilon_3}^2 \quad \sigma_{\epsilon_4}^2 \quad \sigma_{\epsilon_1\epsilon_2} \quad \sigma_{\epsilon_3\epsilon_4}]$$

Notice that repeated parameters (e.g., $\sigma_{\epsilon_3\epsilon_4}$) are not repeated.

Notice that coefficients which are set to zero (e.g., B_{12}) are not included.

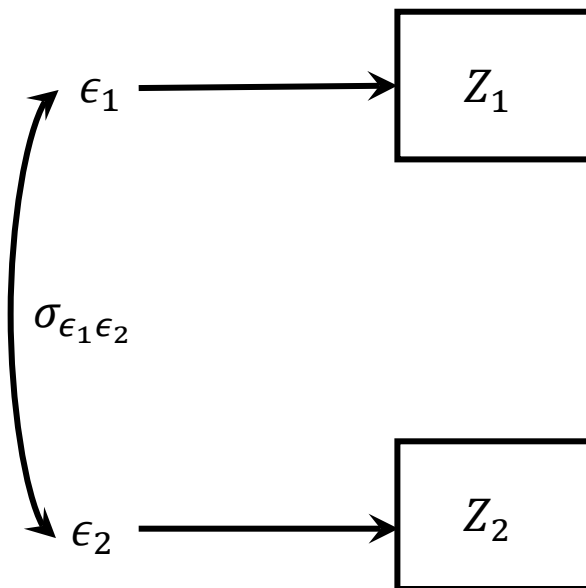
We say that θ_{Σ} has dimensions $t \times 1$

Model Implied Covariance of X s

The model actually just has a matrix used to represent the covariances among X s

$$E(\mathbf{X}\mathbf{X}') = \Sigma_{\epsilon_X\epsilon_X}$$

This means the model implied covariance for the X s is $\Sigma_{\epsilon_X\epsilon_X}$



$$\Sigma_{XX}(\theta) = \Sigma_{\epsilon_X\epsilon_X}$$

What is identification?

If an unknown parameter in θ can be written as a function of one or more elements of Σ_{zz} , that parameter is identified. If all unknown parameters in θ are identified, then the model is identified.

This definition emphasizes how we can actually go about testing whether a model is identified.

Consider two potential parameter vectors: θ_1 and θ_2 . The implied covariance structures for these two parameter vectors are $\Sigma_{zz}(\theta_1)$ and $\Sigma_{zz}(\theta_2)$. The model is identified if when $\Sigma_{zz}(\theta_1) = \Sigma_{zz}(\theta_2)$ means that $\theta_1 = \theta_2$.

No two parameter vectors result in the same implied covariance unless they are equal.

Over vs. Under Identification

Overidentification means there more information than is needed to identify a parameter.



A model is overidentified when all parameters are identified and at least 1 parameter is overidentified

$\sigma_{\epsilon_1\epsilon_2} = \sigma_{Z_1Z_2}$ and $\sigma_{\epsilon_1\epsilon_2} = \sigma_{Z_2Z_3}$ this implies that $\sigma_{Z_1Z_2} = \sigma_{Z_2Z_3}$ in the population, if the model is correct

Underidentification means there is not enough information to identify a parameter.

A model is underidentified when any parameter is underidentified.



$\sigma_{\epsilon_1\epsilon_2} + \sigma_{\epsilon_2\epsilon_3} = \sigma_{Z_1Z_2}$ we cannot individually identify $\sigma_{\epsilon_1\epsilon_2}$ and $\sigma_{\epsilon_2\epsilon_3}$

Why does it matter?

If a model is under-identified, we cannot get estimates of all parameters in the model.

Most of the time this means we need to put restrictions into our model, in order for it to be identified.

Some restrictions the model automatically makes

- The diagonal of the **B** matrix is fixed at zero
- Coefficient matrix for ϵ is an identity matrix

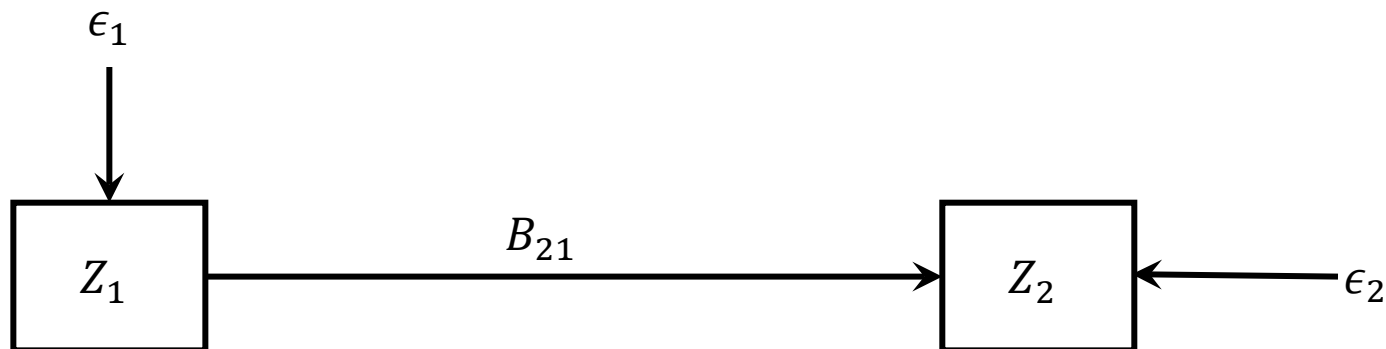
Identification is not related to sample size, larger samples don't help. Identification depends on the number of unique elements in the population covariance matrix.

A simple example

The implied covariance matrix is used to solve for the estimated parameters in the model, based on the observed covariances.

Identification means there is a unique solution for those estimated parameters.

Consider a very simple example... this might look familiar to you!



A simple example

If we have data on Z_1 , and Z_2 , then we would have an observed covariance matrix:

$$\Sigma_{ZZ} = \begin{bmatrix} \sigma_{Z_1}^2 & \sigma_{Z_1 Z_2} \\ \sigma_{Z_1 Z_2} & \sigma_{Z_2}^2 \end{bmatrix}$$

This matrix is symmetric, it has 3 unique elements

In general you will have $\frac{1}{2}(N_Z)(N_Z + 1)$ elements

Here $N_Z = 2$, so $\frac{1}{2}(2)(2 + 1) = \frac{1}{2}(2 * 3) = \frac{6}{2} = 3$

Model Implied Covariance

Our goal is to figure out

$$\Sigma_{ZZ}(\theta) = (I - B)^{-1} \Sigma_{\epsilon\epsilon} (I - B)^{-1'}$$

We need to identify each element based on the model

$$\begin{aligned} Z_1 &= \alpha_1 + \epsilon_1 \\ Z_2 &= \alpha_2 + B_{21}Z_1 + \epsilon_2 \end{aligned}$$

$$\Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & 0 \\ 0 & \sigma_{\epsilon_{Z_2}}^2 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ B_{21} & 0 \end{bmatrix} \quad (\mathbf{I} - \mathbf{B}) = \begin{bmatrix} 1 & 0 \\ -B_{21} & 1 \end{bmatrix}$$

$$(\mathbf{I} - \mathbf{B})^{-1} = \begin{bmatrix} 1 & 0 \\ B_{21} & 1 \end{bmatrix} \quad (\mathbf{I} - \mathbf{B})^{-1'} = \begin{bmatrix} 1 & B_{21} \\ 0 & 1 \end{bmatrix}$$

$$(\mathbf{I} - \mathbf{B})^{-1} \Sigma_{\epsilon\epsilon} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & 0 \\ B_{21} \sigma_{\epsilon_{Z_1}}^2 & \sigma_{\epsilon_{Z_2}}^2 \end{bmatrix}$$

$$(\mathbf{I} - \mathbf{B})^{-1} \Sigma_{\epsilon\epsilon} (\mathbf{I} - \mathbf{B})^{-1'} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & B_{21} \sigma_{\epsilon_{Z_1}}^2 \\ B_{21} \sigma_{\epsilon_{Z_1}}^2 & B_{21}^2 \sigma_{\epsilon_{Z_1}}^2 + \sigma_{\epsilon_{Z_2}}^2 \end{bmatrix}$$

Putting it all together

$$\Sigma = \begin{bmatrix} \sigma_{Z_1}^2 & \sigma_{Z_1 Z_2} \\ \sigma_{Z_1 Z_2} & \sigma_{Z_2}^2 \end{bmatrix}$$

$$\Sigma(\theta) = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & B_{21} \sigma_{\epsilon_{Z_1}}^2 \\ B_{21} \sigma_{\epsilon_{Z_1}}^2 & B_{21}^2 \sigma_{\epsilon_{Z_1}}^2 + \sigma_{\epsilon_{Z_2}}^2 \end{bmatrix}$$

To evaluate whether each parameter is identified we set up a system of equations

$$\sigma_{Z_1}^2 = \sigma_{\epsilon_1}^2$$

This parameter is identified because they are a function of a single known covariance

$$\sigma_{Z_1 Z_2} = B_{21} \sigma_{\epsilon_1}^2$$

$$\sigma_{Z_2}^2 = B_{21}^2 \sigma_{\epsilon_1}^2 + \sigma_{\epsilon_2}^2$$

This looks promising because there are two unknowns B_{21} and $\sigma_{\epsilon_2}^2$ and 2 knowns $\sigma_{Z_1 Z_2}$ and $\sigma_{Z_2}^2$

Is the model identified?

$$\sigma_{Z_1}^2 = \sigma_{\epsilon_1}^2$$

$$\sigma_{Z_1 Z_2} = B_{21} \sigma_{\epsilon_1}^2$$

$$\sigma_{Z_2}^2 = B_{21}^2 \sigma_{\epsilon_1}^2 + \sigma_{\epsilon_2}^2$$

Our goal is to express each parameter as a function of known values

$$\sigma_{\epsilon_1}^2 = \sigma_{Z_1}^2$$

$$B_{21} = \frac{\sigma_{Z_1 Z_2}}{\sigma_{\epsilon_1}^2} = \frac{\sigma_{Z_1 Z_2}}{\sigma_{Z_1}^2}$$

Fun fact: These solution are the same as the solutions for the simple regression model!

$$\sigma_{\epsilon_2}^2 = \sigma_{Z_2}^2 - B_{21}^2 \sigma_{\epsilon_1}^2 = \sigma_{Z_2}^2 - \left(\frac{\sigma_{Z_1 Z_2}}{\sigma_{Z_1}^2} \right)^2 \sigma_{Z_1}^2 = \sigma_{Z_2}^2 - (\sigma_{Z_1 Z_2})^2 / \sigma_{Z_1}^2$$

All of the parameters can be expressed as a function of the covariances, so the model (i.e., every parameter) is identified.

That was A LOT of work, and this is a VERY simple model. So folks have come up with some shortcuts to evaluate whether a model is identified.

Identification Rules

Necessary vs. Sufficient

In mathematics and logic we often use terms to describe the situations under which certain events occur.

In this case we're discussing **identification**

A **necessary** condition for identification means that if it is not fulfilled, the model is not identified.

If A is necessary for I...

Then (not A) implies (not I)

But (A) does not imply (I)

A **sufficient** condition for identification means that if it is fulfilled, the model is identified.

If B is sufficient for I...

Then (B) implies (I)

But (not B) does not imply (not I)

N_θ -Rule

The N_θ -Rule is *necessary but not sufficient* for identification

Applies to

- Observed variable models
- Latent variable models

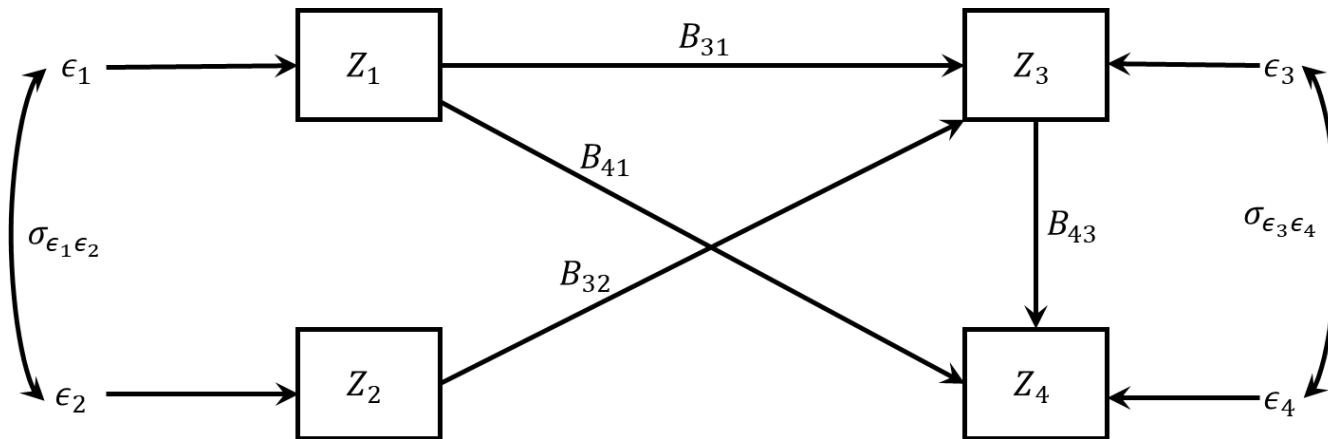
Remember N_θ is the length of θ , a list of all non-repeating and freely estimated parameters (not including intercepts).

We will always have $\frac{1}{2}(N_Z)(N_Z + 1)$ unique elements of the population covariance matrix.

If $N_\theta \leq \frac{1}{2}(N_Z)(N_Z + 1)$ the model *may be identified*

If $N_\theta > \frac{1}{2}(N_Z)(N_Z + 1)$ the model *is not identified*

Applying the N_θ -Rule



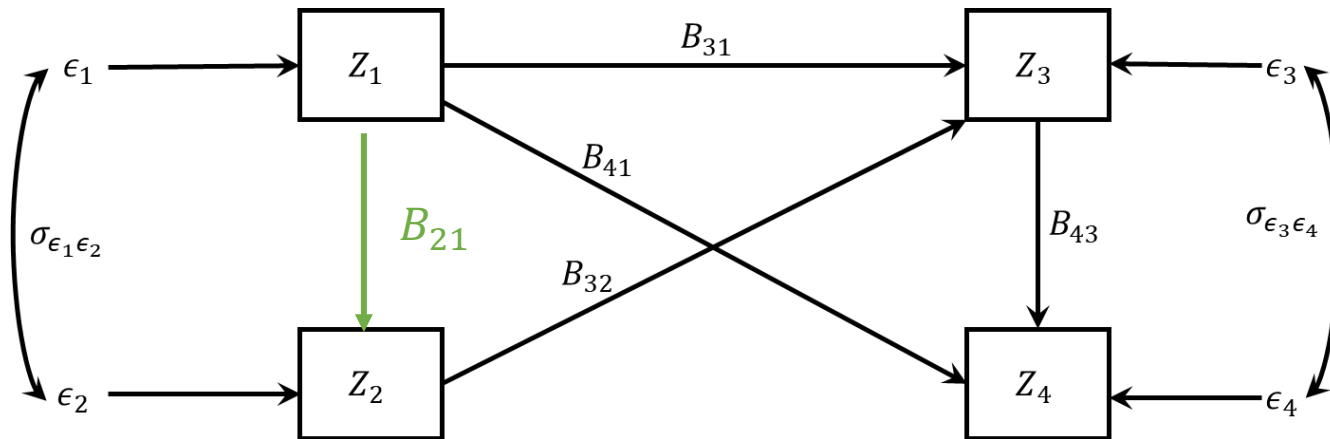
$$\theta_{\Sigma} = [B_{31} \quad B_{32} \quad B_{41} \quad B_{43} \quad \sigma_{\epsilon_1}^2 \quad \sigma_{\epsilon_2}^2 \quad \sigma_{\epsilon_3}^2 \quad \sigma_{\epsilon_4}^2 \quad \sigma_{\epsilon_1\epsilon_2} \quad \sigma_{\epsilon_3\epsilon_4}]$$

How many elements in the covariance matrix? $\frac{1}{2}(4)(5) = \frac{20}{2} = 10$

How many parameters? 10

The model *may* be identified

Applying the N_θ -Rule



$$\theta_{\Sigma} = [B_{21} \quad B_{31} \quad B_{32} \quad B_{41} \quad B_{43} \quad \sigma_{\epsilon_1}^2 \quad \sigma_{\epsilon_2}^2 \quad \sigma_{\epsilon_3}^2 \quad \sigma_{\epsilon_4}^2 \quad \sigma_{\epsilon_1\epsilon_2} \quad \sigma_{\epsilon_3\epsilon_4}]$$

How many elements in the covariance matrix? $\frac{1}{2}(4)(5) = \frac{20}{2} = 10$

How many parameters? 11

The model *is not* identified

Null B_{YY} Rule

The Null B Rule is *sufficient* **not necessary** for identification

Applies to

- Observed variable models

Does not apply to...

- Latent variable models

$$B = \begin{bmatrix} B_{XX} & B_{XY} \\ B_{YX} & B_{YY} \end{bmatrix}$$

B_{XX} and B_{XY} are always **0** because the X variables are always exogenous (nothing predicts them)

If no endogenous variable predicts any other endogenous variable the B_{YY} matrix is **0**

If $B_{YY} = \mathbf{0}$ the model is identified

Why? Null B-Rule

Identification requires expressing the parameters as functions of the known covariances

$$\Sigma = \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY} & \Sigma_{YY} \end{bmatrix}$$

When $B_{YY} = 0$

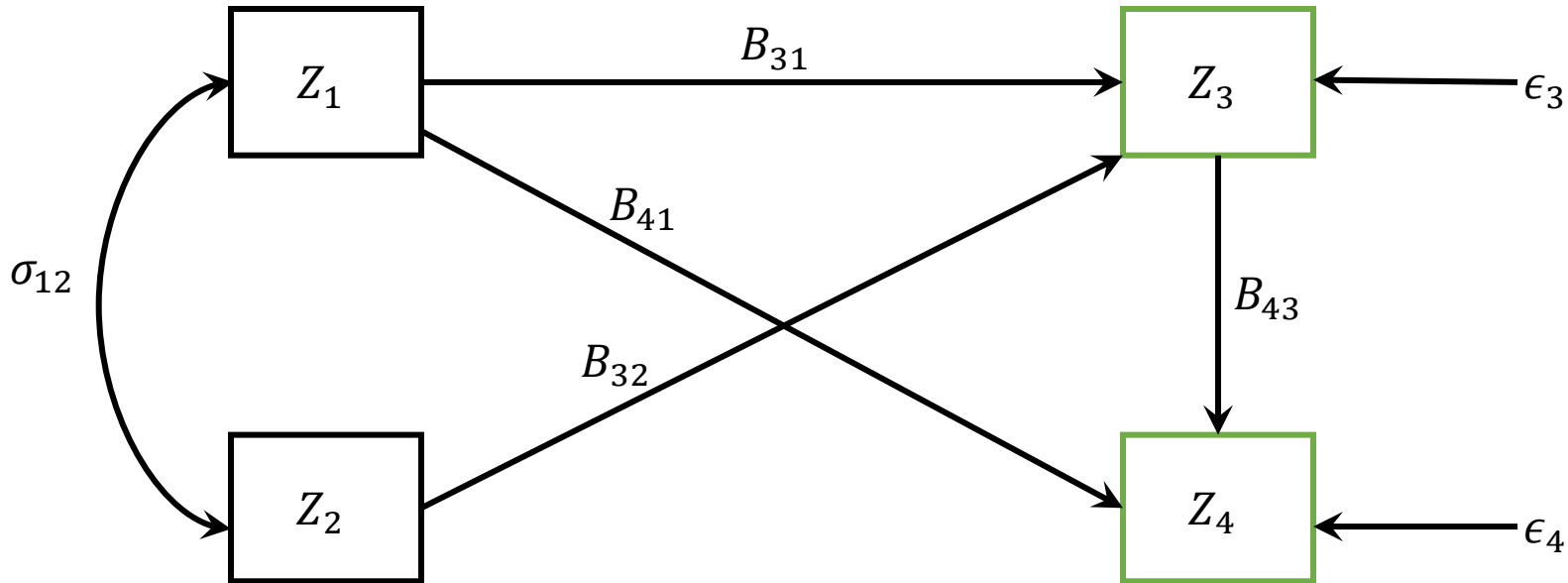
$$\Sigma_{YY} = B_{YX}\Sigma_{XX}B'_{YX} + \Sigma_{\epsilon_Y\epsilon_Y}$$

$$\Sigma_{XY} = \Sigma_{XX}B'_{YX}$$

Σ_{XX} is the known covariance matrix of Xs

$B'_{YX} = \Sigma_{XX}^{-1}\Sigma_{XY}$ so as long as Σ_{XX} is singular B_{YX} is identified

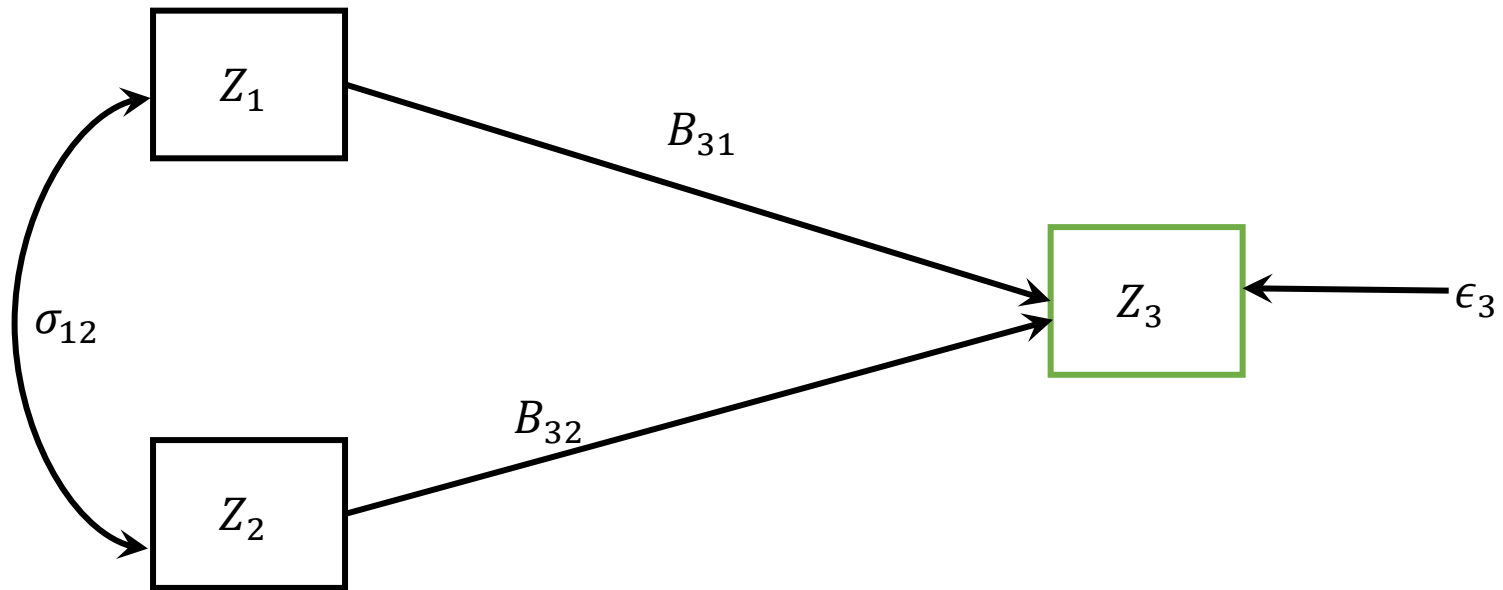
Applying the Null B-Rule



Does the null B rule apply for this model? **NO**

Does this mean the model is not identified? **No, it may still be identified**

Does the Null B Model apply to this model?



Does the null B rule apply for this model? **YES**

Does this mean the model is identified? **YES**

Fully Recursive Rule

The Recursive Rule is *sufficient* ***not necessary*** for identification

Applies to

- Observed variable models

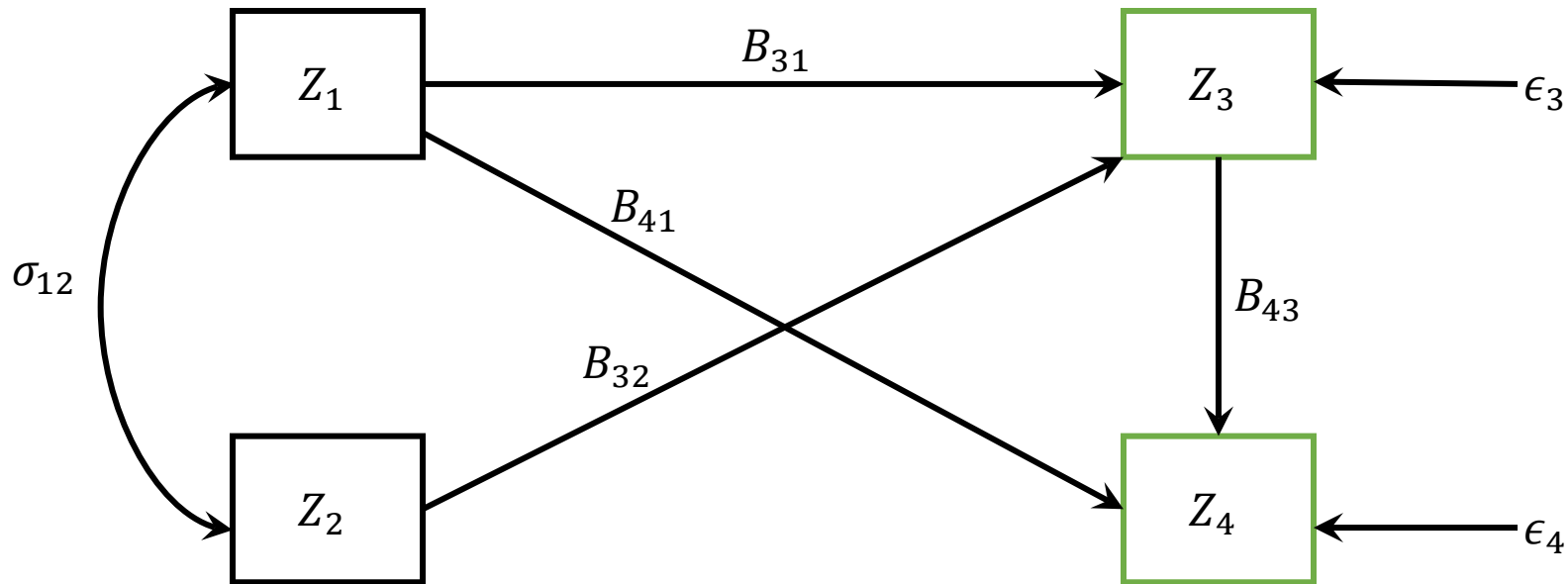
Does not apply to...

- Latent variable models

$\mathbf{B}_{YY}$ matrix must be triangular (upper or lower)

$\Sigma_{\epsilon_Y \epsilon_Y}$ matrix must be diagonal (i.e., no correlated deviations)

Applying the Recursive Rule

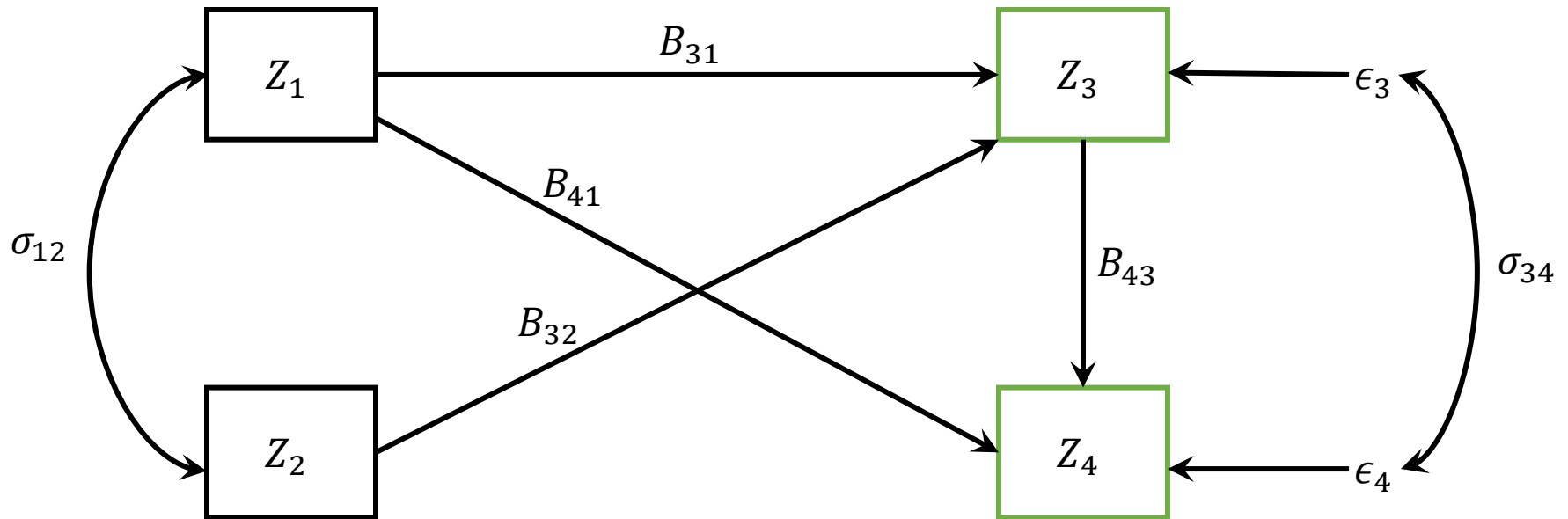


Null B_{YY} rule said this model may be identified

Does the recursive rule apply here? **YES**

Does this mean the model is identified? **Yes**

Applying the Recursive Rule

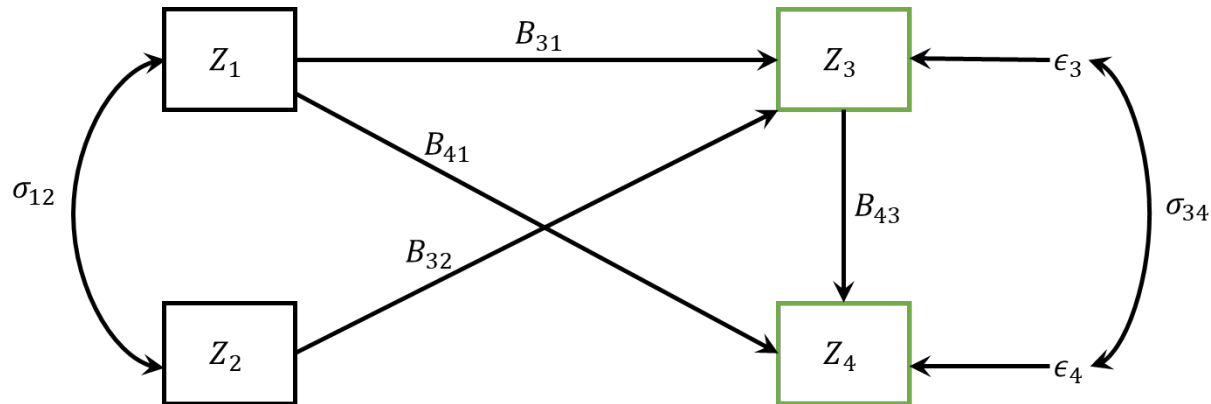


Does the recursive rule apply here? **NO**

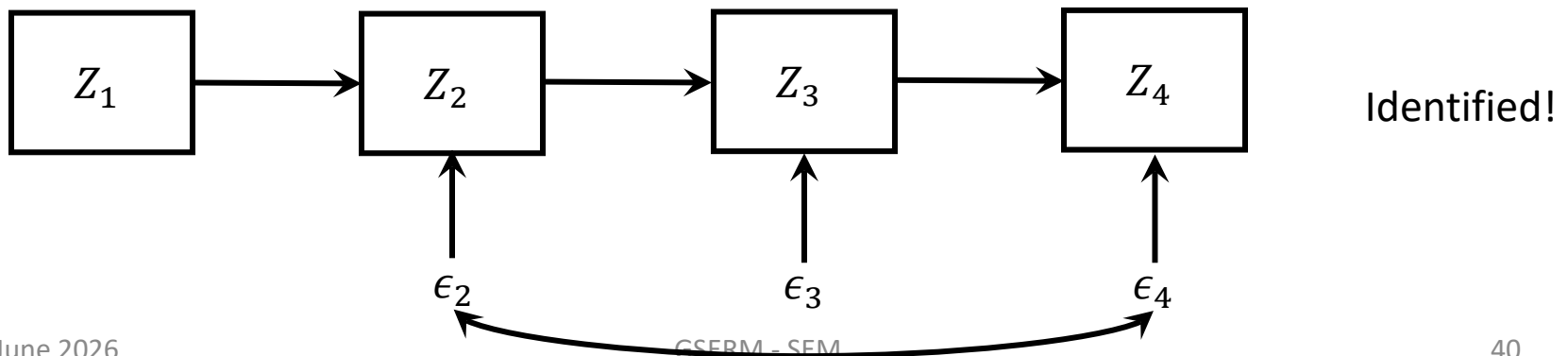
Does this mean the model is not identified? **No, it may still be identified**

Recursive and Correlated Errors Rule

Brito & Pearl (2002) show that as long as disturbances of two *directly connected* variables are uncorrelated, the model is identified



This model is **not identified** because Z_3 and Z_4 are directly connected AND their disturbances are correlated



Matrix Rank

A property of a matrix is its rank, which tells us how much unique information is contained in its columns or rows.

Rank is particularly relevant when we are using matrices to solve systems of equations.

Matrix Rank: The maximum number of linearly independent column vectors (or row vectors)

Coding note:

You can check the rank of a matrix in R with the `Matrix` package and the `rankMatrix` function

Matrix Rank

Consider solving the following system of equations

$$\begin{aligned}1X_1 + 2X_2 - 3X_3 &= 5 \\2X_1 + 4X_2 &= 4 \\X_3 &= -1\end{aligned}$$

$$\begin{bmatrix} 1 & 2 & -3 & 5 \\ 2 & 4 & 0 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

It looks like we have 3 unique pieces of information (equations) so we should be able to solve for the 3 unknowns.

However, some of the information is overlapping.

$$2 \times (\text{Row1}) + 3 \times (\text{Row3}) = \text{Row2}$$

Alternatively,

$$2 \times \text{Col1} = \text{Col2}$$

Matrix Rank

There are only 2 linearly independent vectors (columns or rows) in this matrix, so its rank is 2

$$\begin{bmatrix} 1 & 2 & -3 & 5 \\ 2 & 4 & 0 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$Rank \leq \min(rows, cols)$ The rank is always less than or equal to the smaller of the two: number of rows and number of columns

For square matrices...

If $Rank = rows$ then the matrix is invertible (nonsingular)

If $Rank < rows$ then the matrix is not invertible (singular)

Rank and Order Conditions

The order condition is **necessary** for identification, but the rank condition is **necessary AND sufficient***

Applies to

- Observed variable models
- Anytime $(I - B_{YY})$ is non-singular (i.e., invertible)
 - Particularly useful for non-recursive models

Does not apply to...

- Latent variable models

No restrictions in $\Sigma_{\epsilon_Y \epsilon_Y}$ (i.e., no values are fixed)

Identification is achieved at the equation level; if all equations are identified the model is identified.

Order Conditions

A **necessary** condition for an equation to be identified is that the number of variables **excluded** from the equation be at least $N_Y - 1$. Where N_Y is the number of outcomes.

This rule applies when the only restrictions in the model are exclusions of variables (i.e., setting coefficients to 0 in **B**)

Example: $N_Y = 2$, $N_X = 1$, each equation needs to exclude at least one variable ($2 - 1$)

$$\begin{aligned} Z_2 &= B_{23}Z_3 + \epsilon_2 \\ Z_3 &= B_{31}Z_1 + \epsilon_3 \end{aligned}$$

This model *may* be identified, we don't know yet

Rank Condition

$$\mathbf{C} = [(\mathbf{I} - \mathbf{B}_{YY}) | -\mathbf{B}_{YX}]$$

Delete all columns of $\mathbf{C}$ which do not have zeros in the i th row of $\mathbf{C}$. Call this $\mathbf{C}_i$

A **necessary and sufficient** condition for the identification of the i th equation is that the rank of $\mathbf{C}_i$ equals $(p - 1)$

$$\begin{aligned} Z_2 &= B_{23}Z_3 + \epsilon_2 \\ Z_3 &= B_{31}Z_1 + \epsilon_3 \end{aligned}$$

$$\mathbf{C} = \left[\left(\mathbf{I} - \begin{bmatrix} 0 & B_{23} \\ 0 & 0 \end{bmatrix} \right) \middle| - \begin{bmatrix} 0 \\ B_{31} \end{bmatrix} \right] = \begin{bmatrix} 1 & -B_{23} & 0 \\ 0 & 0 & -B_{31} \end{bmatrix}$$

$$\mathbf{C}_1 = \left[\left(\mathbf{I} - \begin{bmatrix} 0 & B_{23} \\ 0 & 0 \end{bmatrix} \right) \middle| - \begin{bmatrix} 0 \\ B_{31} \end{bmatrix} \right] = \begin{bmatrix} 0 \\ -B_{31} \end{bmatrix}$$

$$\mathbf{C}_2 = \left[\left(\mathbf{I} - \begin{bmatrix} 0 & \beta_{12} \\ 0 & 0 \end{bmatrix} \right) \middle| - \begin{bmatrix} 0 \\ \gamma_{21} \end{bmatrix} \right] = \begin{bmatrix} 1 & -B_{23} \\ 0 & 0 \end{bmatrix}$$

A note on $\Sigma_{\epsilon_Y \epsilon_Y}$

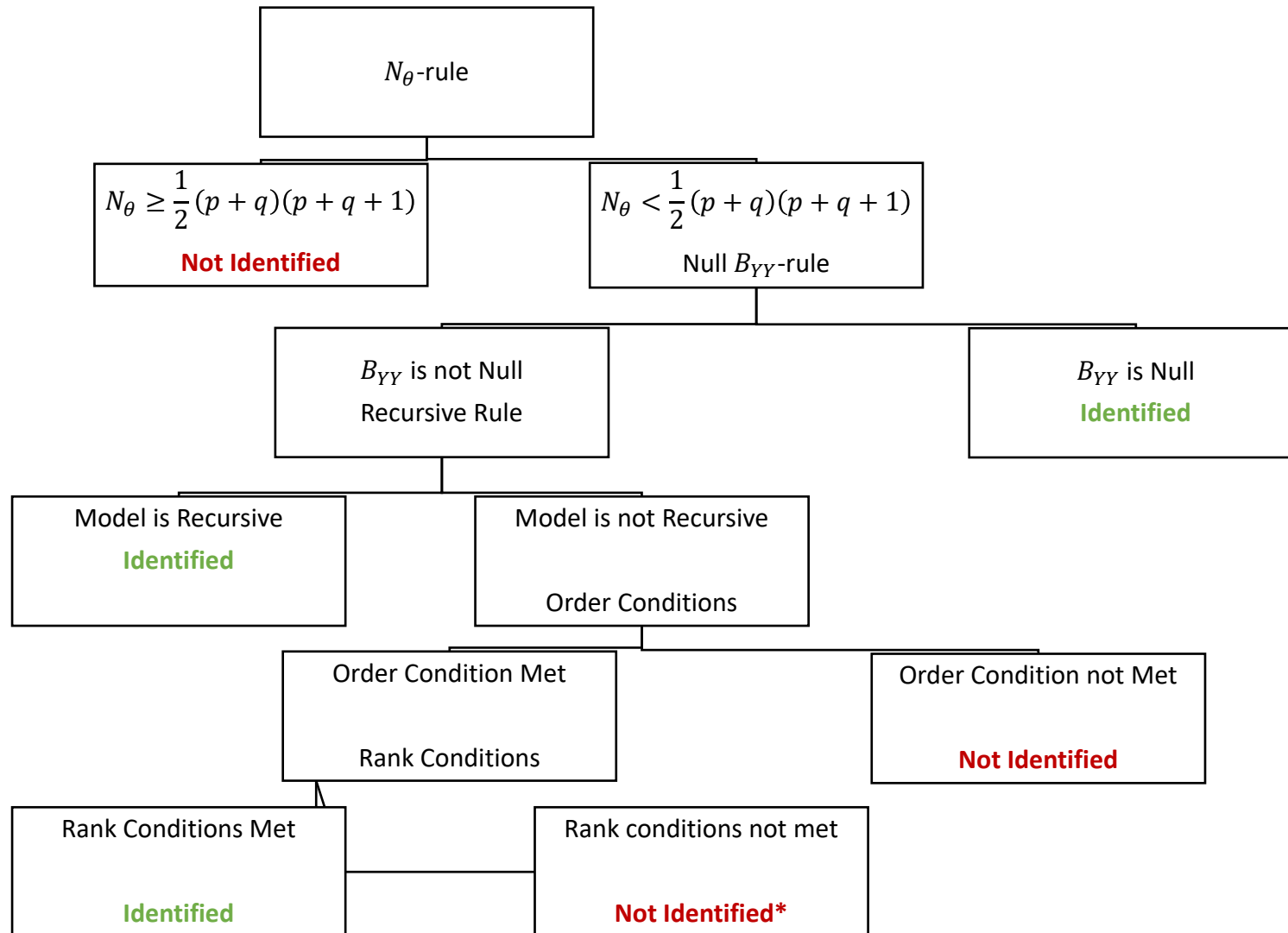
Based on the rank and order conditions, a matrix may not *seem* identified.
But if there are restrictions on $\Sigma_{\epsilon\epsilon}$ it may still be identified.

Rank Rule Assumes:

$$\Sigma_{\epsilon_Y \epsilon_Y} = \begin{bmatrix} \sigma_{Y_1 Y_1}^2 & \cdots & \sigma_{Y_1 Y_{N_Y}} \\ \vdots & \ddots & \vdots \\ \sigma_{Y_1 Y_{N_Y}} & \cdots & \sigma_{Y_{N_Y} Y_{N_Y}}^2 \end{bmatrix}$$

We often assume $\Sigma_{\epsilon_Y \epsilon_Y}$ is diagonal

Identification Rule Summary (Observed Variable Models)



*If additional restrictions are available model may still be identified